

Lab 3: Secure Mail Server Setup and Email Inbox Configuration

Course: Unix-Based Internet Applications

Topic: E-mail and Creation of Secured Mail Server and E-mail Inbox

Duration: 4-6 hours

Difficulty Level: Intermediate to Advanced

Learning Objectives

By the end of this lab work, you will be able to:

1. Understand the architecture of email systems and mail transfer protocols (SMTP, IMAP, POP3)
2. Install and configure Postfix SMTP server for sending emails
3. Install and configure Dovecot IMAP/POP3 server for receiving emails
4. Implement TLS/SSL encryption for secure email transmission
5. Configure email authentication protocols (SPF, DKIM, DMARC)
6. Set up Roundcube webmail interface for email management
7. Test and troubleshoot mail server configurations using command-line tools

Prerequisites

Before starting this lab, you should have:

- Basic Linux command-line knowledge and experience with text editors (nano, vim)
- Understanding of TCP/IP networking fundamentals
- Root or sudo access to an Ubuntu 22.04 LTS or Debian 11 system
- Domain name or ability to configure local DNS for testing
- Familiarity with basic security concepts

Required Software and Tools

- **Operating System:** Ubuntu 22.04 LTS or Debian 11
- **Mail Transfer Agent:** Postfix
- **Mail Delivery Agent:** Dovecot (IMAP/POP3 server)
- **Email Authentication:** OpenDKIM
- **Web Server:** Nginx or Apache
- **Database:** MariaDB or MySQL

- **Webmail Client:** Roundcube
- **SSL/TLS Certificates:** Let's Encrypt (certbot) or self-signed certificates
- **Testing Tools:** telnet, openssl, mail utilities

Lab Architecture Overview

The mail server architecture consists of the following components:

- **Postfix:** Handles outgoing email (SMTP) and routing
- **Dovecot:** Manages incoming email access (IMAP/POP3)
- **OpenDKIM:** Signs outgoing emails with DKIM signatures
- **Roundcube:** Provides web-based email access
- **TLS/SSL:** Encrypts communications between clients and servers

Port Configuration:

- Port 25: SMTP (server-to-server)
- Port 587: Submission (client-to-server with STARTTLS)
- Port 465: SMTPS (deprecated but sometimes used)
- Port 143: IMAP (unencrypted)
- Port 993: IMAPS (IMAP with SSL/TLS)
- Port 110: POP3 (unencrypted)
- Port 995: POP3S (POP3 with SSL/TLS)

Part 1: System Preparation and Initial Setup

Task 1.1: Update System Packages

Update your system to ensure all packages are current:

```
sudo apt update
sudo apt upgrade -y
```

Task 1.2: Configure Hostname

Set your system hostname (replace `mail.example.com` with your domain):

```
sudo hostnamectl set-hostname mail.example.com
```

Edit `/etc/hosts` to include your hostname:

```
sudo nano /etc/hosts
```

Add the following line (replace with your IP):

```
192.168.1.100 mail.example.com mail
```

Task 1.3: Configure DNS Records (if using a real domain)

Configure the following DNS records for your domain:

- **A Record:** mail.example.com → Your server IP
- **MX Record:** example.com → mail.example.com (Priority: 10)
- **PTR Record:** Reverse DNS pointing to mail.example.com

For testing purposes, you can skip this and use local configuration.

Task 1.4: Install Essential Tools

```
sudo apt install -y curl wget git net-tools dnsutils telnet
```

Deliverable 1.1: Take a screenshot showing the output of `hostname -f` and `cat /etc/hosts`.

Part 2: Postfix SMTP Server Installation and Configuration

Task 2.1: Install Postfix

Install Postfix mail server:

```
sudo apt install -y postfix
```

During installation:

- Select **"Internet Site"** as the configuration type
- Enter your domain name (e.g., example.com)

Task 2.2: Configure Postfix Main Configuration

Edit the main Postfix configuration file:

```
sudo nano /etc/postfix/main.cf
```

Add or modify the following parameters:

```
# Basic Settings
myhostname = mail.example.com
mydomain = example.com
myorigin = $mydomain
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain

# Network Settings
inet_interfaces = all
inet_protocols = ipv4

# Mail Storage
home_mailbox = Maildir/
mailbox_size_limit = 0
recipient_delimiter = +

# Virtual Mailbox Settings (for virtual users)
virtual_mailbox_domains = example.com
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
virtual_gid_maps = static:5000

# SMTP Client Restrictions
smtpd_relay_restrictions = permit_mynetworks,
    permit_sasl_authenticated,
    defer_unauth_destination
```

Task 2.3: Configure SMTP Authentication

Add SASL authentication settings to `/etc/postfix/main.cf`:

```
# SMTP Authentication
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
broken_sasl_auth_clients = yes
```

Task 2.4: Configure Master Process

Edit `/etc/postfix/master.cf` to enable submission port:

```
sudo nano /etc/postfix/master.cf
```

Uncomment and configure the submission service (port 587):

```
submission inet n      -      y      -      -      smtpd
  -o syslog_name=postfix/submission
```

```
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
-o smtpd_reject_unlisted_recipient=no
-o smtpd_client_restrictions=permit_sasl_authenticated,reject
-o smtpd_relay_restrictions=permit_sasl_authenticated,reject
```

Task 2.5: Create Virtual Mailbox Map

Create the virtual mailbox mapping file:

```
sudo nano /etc/postfix/vmailbox
```

Add user mappings:

```
user1@example.com    example.com/user1/
user2@example.com    example.com/user2/
```

Generate the hash database:

```
sudo postmap /etc/postfix/vmailbox
```

Task 2.6: Create Mail Directory and User

```
sudo groupadd -g 5000 vmail
sudo useradd -g vmail -u 5000 vmail -d /var/mail/vhosts -m
sudo mkdir -p /var/mail/vhosts/example.com
sudo chown -R vmail:vmail /var/mail/vhosts
```

Task 2.7: Restart Postfix

```
sudo systemctl restart postfix
sudo systemctl enable postfix
sudo systemctl status postfix
```

Deliverable 2.1: Provide the output of `postconf -n` command.

Deliverable 2.2: Take a screenshot showing Postfix service status.

Part 3: Dovecot IMAP/POP3 Server Setup

Task 3.1: Install Dovecot

```
sudo apt install -y dovecot-core dovecot-imapd dovecot-pop3d
```

Task 3.2: Configure Dovecot Main Settings

Edit /etc/dovecot/dovecot.conf:

```
sudo nano /etc/dovecot/dovecot.conf
```

Ensure the following line is uncommented:

```
protocols = imap pop3 lmtp  
listen = *, ::
```

Task 3.3: Configure Mail Location

Edit /etc/dovecot/conf.d/10-mail.conf:

```
sudo nano /etc/dovecot/conf.d/10-mail.conf
```

Configure mail location:

```
mail_location = maildir:/var/mail/vhosts/%d/%n  
mail_privileged_group = mail  
  
namespace inbox {  
    inbox = yes  
}
```

Task 3.4: Configure Authentication

Edit /etc/dovecot/conf.d/10-auth.conf:

```
sudo nano /etc/dovecot/conf.d/10-auth.conf
```

Modify the following:

```
disable_plaintext_auth = yes  
auth_mechanisms = plain login
```

Comment out the system auth include and uncomment:

```
!include auth-passwdfile.conf.ext
```

Task 3.5: Configure User Database

Edit `/etc/dovecot/conf.d/auth-passwdfile.conf.ext`:

```
sudo nano /etc/dovecot/conf.d/auth-passwdfile.conf.ext
```

Configure:

```
passdb {  
    driver = passwd-file  
    args = scheme=PLAIN username_format=%u /etc/dovecot/users  
}  
  
userdb {  
    driver = static  
    args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n  
}
```

Create the users file:

```
sudo nano /etc/dovecot/users
```

Add users (format: `username:password`):

```
user1@example.com:{PLAIN}password123  
user2@example.com:{PLAIN}password456
```

Set permissions:

```
sudo chmod 640 /etc/dovecot/users  
sudo chown root:dovecot /etc/dovecot/users
```

Task 3.6: Configure Dovecot SASL for Postfix

Edit `/etc/dovecot/conf.d/10-master.conf`:

```
sudo nano /etc/dovecot/conf.d/10-master.conf
```

Find the `service auth` section and add:

```
service auth {  
    unix_listener /var/spool/postfix/private/auth {  
        mode = 0660  
        user = postfix  
        group = postfix  
    }  
}
```

```
}  
}
```

Task 3.7: Restart Dovecot

```
sudo systemctl restart dovecot  
sudo systemctl enable dovecot  
sudo systemctl status dovecot
```

Deliverable 3.1: Provide the output of `doveconf -n` command.

Deliverable 3.2: Verify Dovecot is listening on ports 143 and 110 using `ss -tlnp | grep dovecot`.

Part 4: Security Implementation - TLS/SSL Encryption

Task 4.1: Generate Self-Signed SSL Certificate (for testing)

```
sudo mkdir -p /etc/ssl/mail  
cd /etc/ssl/mail  
sudo openssl req -new -x509 -days 365 -nodes \  
-out mail.example.com.crt \  
-keyout mail.example.com.key
```

Fill in the certificate details when prompted.

Set proper permissions:

```
sudo chmod 600 /etc/ssl/mail/mail.example.com.key  
sudo chmod 644 /etc/ssl/mail/mail.example.com.crt
```

Note: For production environments, use Let's Encrypt:

```
sudo apt install -y certbot  
sudo certbot certonly --standalone -d mail.example.com
```

Task 4.2: Configure Postfix TLS

Edit `/etc/postfix/main.cf` and add:

```
# TLS Settings  
smtpd_tls_cert_file=/etc/ssl/mail/mail.example.com.crt  
smtpd_tls_key_file=/etc/ssl/mail/mail.example.com.key  
smtpd_use_tls=yes  
smtpd_tls_security_level=may  
smtpd_tls_auth_only = yes  
smtpd_tls_loglevel = 1
```

```
smtpd_tls_received_header = yes
smtpd_tls_session_cache_timeout = 3600s

# Outgoing TLS
smtp_tls_security_level = may
smtp_tls_loglevel = 1
```

Task 4.3: Configure Dovecot SSL

Edit /etc/dovecot/conf.d/10-ssl.conf:

```
sudo nano /etc/dovecot/conf.d/10-ssl.conf
```

Configure SSL settings:

```
ssl = required
ssl_cert = &lt;/etc/ssl/mail/mail.example.com.crt
ssl_key = &lt;/etc/ssl/mail/mail.example.com.key
ssl_min_protocol = TLSv1.2
ssl_prefer_server_ciphers = yes
```

Task 4.4: Restart Services

```
sudo systemctl restart postfix
sudo systemctl restart dovecot
```

Task 4.5: Configure Firewall

Allow necessary mail server ports:

```
sudo ufw allow 25/tcp
sudo ufw allow 587/tcp
sudo ufw allow 465/tcp
sudo ufw allow 143/tcp
sudo ufw allow 993/tcp
sudo ufw allow 110/tcp
sudo ufw allow 995/tcp
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw enable
```

Deliverable 4.1: Test TLS connection using: `openssl s_client -connect localhost:587 -starttls smtp`

Deliverable 4.2: Provide screenshot showing successful TLS handshake.

Part 5: Email Authentication Configuration

Task 5.1: Configure SPF Record

Add a TXT record to your DNS:

```
example.com.  IN  TXT  "v=spf1 mx a ip4:YOUR_SERVER_IP -all"
```

This tells receiving servers that only your mail server can send emails from your domain.

Task 5.2: Install OpenDKIM

```
sudo apt install -y opendkim opendkim-tools
```

Task 5.3: Configure OpenDKIM

Edit `/etc/opendkim.conf`:

```
sudo nano /etc/opendkim.conf
```

Configure:

Domain	example.com
KeyFile	/etc/opendkim/keys/example.com/default.private
Selector	default
Socket	inet:8891@localhost
Mode	sv
Syslog	yes
LogWhy	yes
Canonicalization	relaxed/simple

Task 5.4: Generate DKIM Keys

```
sudo mkdir -p /etc/opendkim/keys/example.com
sudo opendkim-genkey -D /etc/opendkim/keys/example.com -d example.com -s default
sudo chown -R opendkim:opendkim /etc/opendkim/keys/
```

View the public key:

```
sudo cat /etc/opendkim/keys/example.com/default.txt
```

Task 5.5: Add DKIM DNS Record

Add a TXT record with the content from `default.txt`:

```
default._domainkey.example.com.  IN  TXT  "v=DKIM1; k=rsa; p=YOUR_PUBLIC_KEY"
```

Task 5.6: Connect OpenDKIM to Postfix

Edit `/etc/postfix/main.cf` and add:

```
# DKIM
milter_default_action = accept
milter_protocol = 6
smtpd_milters = inet:localhost:8891
non_smtpd_milters = inet:localhost:8891
```

Task 5.7: Configure DMARC Policy

Add a TXT record for DMARC:

```
_dmarc.example.com.  IN  TXT  "v=DMARC1; p=quarantine; rua=mailto:postmaster@example.com"
```

Task 5.8: Restart Services

```
sudo systemctl restart opendkim
sudo systemctl restart postfix
```

Deliverable 5.1: Provide the DKIM public key from `default.txt`.

Deliverable 5.2: Show OpenDKIM service status.

Part 6: Webmail Interface Setup (Roundcube)

Task 6.1: Install Web Server and PHP

```
sudo apt install -y nginx php-fpm php-cli php-imap php-json \
  php-mysql php-mbstring php-xml php-zip php-intl \
  php-imagick php-curl php-gd
```

Task 6.2: Install MariaDB

```
sudo apt install -y mariadb-server
sudo mysql_secure_installation
```

Task 6.3: Create Database for Roundcube

```
sudo mysql -u root -p
```

Run the following SQL commands:

```
CREATE DATABASE roundcube CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'roundcube'@'localhost' IDENTIFIED BY 'strong_password';
GRANT ALL PRIVILEGES ON roundcube.* TO 'roundcube'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Task 6.4: Download and Install Roundcube

```
cd /tmp
wget https://github.com/roundcube/roundcubemail/releases/download/1.6.5/roundcubemail-1.6.5-complete.tar.gz
tar -xzf roundcubemail-1.6.5-complete.tar.gz
sudo mv roundcubemail-1.6.5 /var/www/roundcube
sudo chown -R www-data:www-data /var/www/roundcube
```

Task 6.5: Configure Nginx for Roundcube

Create Nginx configuration:

```
sudo nano /etc/nginx/sites-available/roundcube
```

Add:

```
server {
    listen 80;
    server_name mail.example.com;
    root /var/www/roundcube;
    index index.php;

    location / {
        try_files $uri $uri/ /index.php;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/var/run/php/php8.1-fpm.sock;
    }
}
```

```
location ~ /\.ht {  
    deny all;  
}  
}
```

Enable the site:

```
sudo ln -s /etc/nginx/sites-available/roundcube /etc/nginx/sites-enabled/  
sudo nginx -t  
sudo systemctl restart nginx
```

Task 6.6: Configure Roundcube

Visit <http://mail.example.com/installer> in your browser to complete the installation.

Database setup:

- Database type: MySQL
- Database name: roundcube
- Database user: roundcube
- Database password: (your password)

IMAP settings:

- IMAP server: ssl://localhost:993

SMTP settings:

- SMTP server: tls://localhost:587
- Use authentication: Yes

After configuration, download the `config.inc.php` file and place it in `/var/www/roundcube/config/`.

Remove the installer for security:

```
sudo rm -rf /var/www/roundcube/installer
```

Deliverable 6.1: Take a screenshot of the Roundcube login page.

Deliverable 6.2: Screenshot showing successful email access through Roundcube.

Part 7: Testing and Troubleshooting

Task 7.1: Test SMTP with Telnet

Test basic SMTP connectivity:

```
telnet localhost 25
```

Commands to execute:

```
EHLO localhost
MAIL FROM: <user1@example.com>
RCPT TO: <user2@example.com>
DATA
Subject: Test Email
This is a test email.
.
QUIT
```

Task 7.2: Test SMTP with TLS

```
openssl s_client -connect localhost:587 -starttls smtp
```

After connection, test SMTP commands.

Task 7.3: Test IMAP Connection

```
openssl s_client -connect localhost:993
```

Commands:

```
a001 LOGIN user1@example.com password123
a002 LIST "" "*"
a003 SELECT INBOX
a004 LOGOUT
```

Task 7.4: Send Test Email

```
echo "Test email body" | mail -s "Test Subject" user2@example.com
```

Task 7.5: Check Mail Logs

View Postfix logs:

```
sudo tail -f /var/log/mail.log
```

Or on some systems:

```
sudo tail -f /var/log/maillog
```

Task 7.6: Verify Email Authentication

Send an email to an external address (Gmail, etc.) and check headers for:

- SPF: PASS
- DKIM: PASS
- DMARC: PASS

Task 7.7: Test Mail Queue

Check mail queue:

```
mailq  
postqueue -p
```

Flush queue:

```
postqueue -f
```

Task 7.8: Common Troubleshooting Commands

Check Postfix configuration:

```
postconf -n
```

Check Dovecot configuration:

```
doveconf -n
```

Check service status:

```
systemctl status postfix  
systemctl status dovecot  
systemctl status opendkim
```

Check listening ports:

```
ss -tlnp | grep -E ':(25|587|465|143|993|110|995)'
```

Deliverable 7.1: Provide complete mail.log output from sending a test email.

Deliverable 7.2: Screenshot showing email headers with SPF, DKIM, and DMARC results.

Deliverable 7.3: Demonstrate successful email send/receive through Roundcube.

Lab Report Requirements

Your lab report should include the following sections:

1. Title Page

- Lab title
- Your name and student ID
- Date of submission
- Course name and instructor

2. Abstract

A brief summary (150-250 words) of what was accomplished in the lab.

3. Introduction

- Purpose of the lab
- Overview of email server architecture
- Brief description of protocols used (SMTP, IMAP, POP3)

4. Methodology

Describe step-by-step what you did in each part of the lab. Include:

- Installation commands
- Configuration file modifications
- Any problems encountered and solutions

5. Results

Present all deliverables requested throughout the lab:

- Screenshots
- Configuration file outputs
- Test results
- Log file excerpts

6. Security Analysis

Discuss the security measures implemented:

- TLS/SSL encryption
- Authentication mechanisms (SPF, DKIM, DMARC)
- Port security and firewall configuration
- Best practices followed

7. Discussion

- Challenges faced during implementation
- How email security protocols work together
- Comparison of different mail server components
- Real-world applications and use cases

8. Conclusion

- Summary of what was learned
- Reflection on mail server security
- Suggestions for further improvements

9. References

List all resources, documentation, and articles consulted.

Evaluation Criteria

Your lab work will be evaluated based on:

Criterion	Points	Description
Postfix Configuration	15	Correct installation and configuration of SMTP server
Dovecot Configuration	15	Proper setup of IMAP/POP3 server
TLS/SSL Implementation	15	Successful encryption configuration
Email Authentication	15	SPF, DKIM, DMARC setup
Roundcube Installation	10	Working webmail interface
Testing & Troubleshooting	10	Comprehensive testing and log analysis
Lab Report Quality	10	Clear documentation and analysis
Screenshots & Deliverables	10	All required evidence provided
Total	100	

Additional Resources

Documentation

- Postfix Documentation: <http://www.postfix.org/documentation.html>
- Dovecot Wiki: <https://doc.dovecot.org/>
- OpenDKIM Documentation: <http://www.opendkim.org/>
- Roundcube Documentation: <https://github.com/roundcube/roundcubemail/wiki>

Useful Commands Reference

Postfix:

```
postconf -n          # Show non-default settings
postfix check         # Check configuration
postqueue -p         # Show mail queue
postsuper -d ALL      # Delete all queued mail
```

Dovecot:

```
doveconf -n          # Show non-default settings
doveadm user &lt;user> # Show user info
doveadm log find      # Find log files
```

Mail Testing:

```
telnet localhost 25 # Test SMTP
openssl s_client -connect localhost:587 -starttls smtp
openssl s_client -connect localhost:993
mail -s "Subject" user@example.com &lt; /path/to/message.txt
```

Log Monitoring:

```
tail -f /var/log/mail.log
journalctl -u postfix -f
journalctl -u dovecot -f
```

Security Best Practices Checklist

- ☐ Changed default passwords for all accounts
- ☐ Enabled TLS/SSL for all connections
- ☐ Configured SPF, DKIM, and DMARC
- ☐ Restricted SMTP relay to authenticated users only
- ☐ Configured firewall to allow only necessary ports

- ☐ Implemented rate limiting to prevent spam
- ☐ Set up reverse DNS (PTR record)
- ☐ Disabled plaintext authentication
- ☐ Regular security updates applied
- ☐ Monitoring and logging enabled
- ☐ Backup procedures established

Submission Guidelines

1. Complete all seven parts of the lab
2. Collect all required deliverables (screenshots, configurations, outputs)
3. Write a comprehensive lab report following the structure outlined
4. Submit the following files:
 - Lab report (PDF format)
 - Configuration files (main.cf, dovecot.conf, etc.)

Submission Deadline: [december 6]

Submission Method: [Via Moodle]

Frequently Asked Questions

Q1: Can I use a different Linux distribution?

A: Yes, but commands may vary slightly. This lab is designed for Ubuntu/Debian. For RHEL/CentOS, use yum or dnf instead of apt.

Q2: What if I don't have a real domain name?

A: You can use a local domain for testing purposes. Configure `/etc/hosts` on your client machines to point to your mail server IP.

Q3: Can I skip the webmail installation?

A: Roundcube installation is part of the lab requirements, but you can use any desktop email client (Thunderbird, Evolution) for testing.

Q4: How do I reset if something goes wrong?

A: You can reinstall packages or restore configuration files from backups. Always keep backup copies before making major changes.

Q5: Is this configuration production-ready?

A: This lab provides a foundation, but production servers require additional hardening, monitoring, spam filtering, and regular maintenance.

